

AC Power Calculations:

$$\theta = \alpha - \beta = \angle \bar{V} - \angle \bar{I}$$

$$\bar{S} = P + jQ = S \angle \theta = (VI) \angle \theta = \bar{V} \cdot \bar{I}^*$$

$$pf = \cos \theta \quad \text{lagging if } \theta > 0, \quad \text{leading if } \theta < 0, \quad \text{unity if } \theta = 0$$

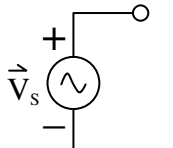
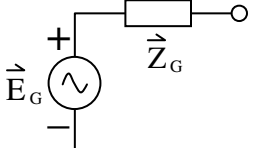
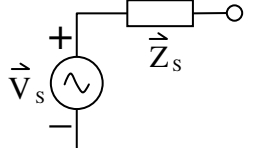
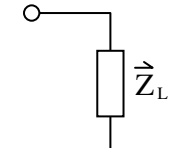
$$\bar{S}_{3\phi} = P_{3\phi} + jQ_{3\phi} = S_{3\phi} \angle \theta = (3V_{LN} I_L) \angle \theta = (\sqrt{3} V_{LL} I_L) \angle \theta = 3 \cdot \bar{S}_{LN}$$

Efficiency Calculations:

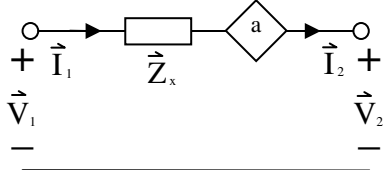
$$\eta_{TL} \text{ or } \eta_{XFMR} = (P_{OUT} / P_{IN}) \times 100,$$

$$\eta_{SYS} = (P_{LNET} / P_{SNET}) \times 100$$

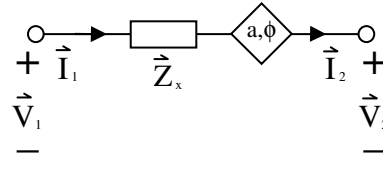
Source and Load Modeling:

Ideal Source Model	Generator Model	Utility Tie Model	Load Model
			
- divide $\bar{Z}_G$ or $\bar{Z}_L$ by 3 if a 3ϕ generator or 3ϕ load is Δ-connected - for utility ties: $Z_s = \frac{V_{rated}^2}{S_{SC}}$ or $Z_s = \frac{V_{rated}}{I_{SC}}$ or $Z_s = \frac{V_{LLrated}^2}{S_{3\phi SC}}$ or $Z_s = \frac{V_{LLrated}}{\sqrt{3} \cdot I_{LSC}}$ and $\theta_s = \tan^{-1}(X/R)$			

Single Phase Power Transformer Equivalent Circuit and Input/Output Equations

	$\begin{cases} a = TTR = V_{1rated} / V_{2rated} \\ \bar{Z}_x = R_1 + jX_1 \\ = (r_1 + jx_1) + a^2 (r_2 + jx_2) \end{cases}$
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Three Phase Power Transformer Per Phase Equivalent Circuit and Input/Output Equations

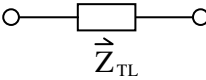
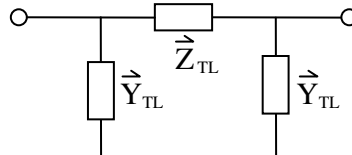
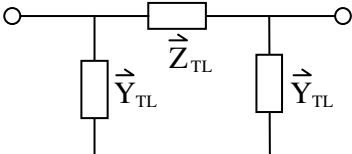
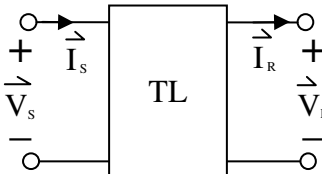
	$\begin{cases} a = TTR = V_{LL1rated} / V_{LL2rated} \\ \bar{Z}_x = R_1 + jX_1 \\ = (r_1 + jx_1) + a^2 (r_2 + jx_2) \\ \text{divide } (r + jx) \text{ value by 3} \\ \text{if the side is } \Delta\text{-connected} \\ \phi = 0^\circ \text{ if Y-Y or } \Delta\text{-}\Delta \text{ or } \Delta_{CBA}\text{-}\Delta_{CBA} \\ = -30^\circ \text{ if Y-}\Delta \text{ or } \Delta_{CBA}\text{-Y} \\ = +30^\circ \text{ if Y-}\Delta_{CBA} \text{ or } \Delta\text{-Y} \\ = -60^\circ \text{ if } \Delta_{CBA}\text{-}\Delta \\ = +60^\circ \text{ if } \Delta\text{-}\Delta_{CBA} \end{cases}$
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$$\bar{Z}_x \text{ in } \Omega = \left(\frac{R_1 + jX_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right]$$

$$\bar{Z}_x \text{ in } \Omega = \left(\frac{Z_x \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \angle \theta_x$$

$$\bar{Z}_x \text{ in } \Omega = \left(\frac{r_1 + jx_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \text{ in } \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$$

divide $(r + jx)$ by 3 if the side is Δ

Transmission Line Equivalent Circuit Models		
Distributed Series Impedance : $\vec{\mathcal{Z}} = R + j\omega L$ ($\Omega / \text{length unit}$)		Distributed Shunt Admittance : $\vec{y} = G + j\omega C$ ($\mathcal{U} / \text{length unit}$)
Propagation Constant : $\vec{\gamma} = \sqrt{\vec{\mathcal{Z}} \cdot \vec{y}}$ ($1 / \text{length unit}$)		Characteristic Impedance : $\vec{Z}_c = \sqrt{\vec{\mathcal{Z}} / \vec{y}}$ (Ω)
Series Impedance Model  $\vec{Z}_{TL} = \vec{\mathcal{Z}} \ell$ Pi Model  $\vec{Y}_{TL} = \frac{1}{2} \vec{y} \ell$		Long Length Model  $\vec{Z}_{TL} = \vec{Z}_c \sinh(\vec{\gamma} \ell)$ $\vec{Y}_{TL} = \frac{\cosh(\vec{\gamma} \ell) - 1}{\vec{Z}_c \sinh(\vec{\gamma} \ell)}$
Single Phase Transmission Line Impedance Parameter Calculations		
$R = \left(\frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y} \right)$	$L = k_{L1} \cdot \ln \left(\frac{GMD_{xy}}{\sqrt{GMR_x \cdot GMR_y}} \right)$	$C = k_{C1} / \ln \left(\frac{GMD_{xy}}{\sqrt{mGMR_x \cdot mGMR_y}} \right)$
$GMD_{xy} = \left[\prod_{i=1}^{n_x} \left(\prod_{j=1}^{n_y} D_{xijy} \right) \right]^{\frac{1}{n_x n_y}}$	$k_{L1} = 6.437376 \times 10^{-4} H / \text{mile}$	$k_{C1} = 4.47659 \times 10^{-8} F / \text{mile}$
Three Phase Transmission Line Impedance Parameter Calculations		
$R = \left(\frac{R_s}{n} \right)$	$L = k_{L3} \cdot \ln \left(\frac{GMD_{eq}}{GMR} \right)$	$C = k_{C3} / \ln \left(\frac{GMD_{eq}}{mGMR} \right)$
$GMD_{eq} = \sqrt[3]{D_{AB} D_{BC} D_{CA}}$	$k_{L3} = 3.218688 \times 10^{-4} H / \text{mile}$	$k_{C3} = 8.953181 \times 10^{-8} F / \text{mile}$
Single Phase and Three Phase Transmission Line Conductor GMR Calculations		
Single Cable (no bundling)	$GMR = GMR_s$	$mGMR = r_s$
Standard 2-Cable Bundle	$GMR = \sqrt{d \cdot GMR_s}$	$mGMR = \sqrt{d \cdot r_s}$
Standard 3-Cable Equilateral Bundle	$GMR = \sqrt[3]{d^2 \cdot GMR_s}$	$mGMR = \sqrt[3]{d^2 \cdot r_s}$
Standard 4-Cable Square Bundle	$GMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot GMR_s}$	$mGMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot r_s}$
For all Non-Standard Bundles	$GMR = \left[\prod_{i=1}^n \left(\prod_{j=1}^n D_{ij} \right) \right]^{\frac{1}{n^2}} \qquad mGMR = \left[\prod_{i=1}^n \left(\prod_{j=1}^n D_{ij}^* \right) \right]^{\frac{1}{n^2}}$ $D_{ij} = D_{ij}^* = \text{distance between subconductor } i \text{ and subconductor } j$ $D_{ii} = GMR_s, \quad D_{ii}^* = r_s \text{ for both stranded and solid cables}$ $GMR_s = \text{table } GMR_s \text{ for stranded cables, or } GMR_s = e^{-0.25} r_s \text{ if solid cables}$	
Transmission Line 2-Port Network Modeling and Equations		
Sending End  Receiving End	Forward Equations $\begin{cases} \vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R \\ \vec{I}_S = \vec{C} \vec{V}_R + \vec{D} \vec{I}_R \end{cases}$	Reverse Equations $\begin{cases} \vec{V}_R = \vec{A} \vec{V}_S - \vec{B} \vec{I}_S \\ \vec{I}_R = -\vec{C} \vec{V}_S + \vec{D} \vec{I}_S \end{cases}$
Short Length (Series Impedance) Model	$\vec{A} = \vec{D} = 1, \quad \vec{B} = \vec{Z}_{TL}, \quad \vec{C} = 0$	
Medium Length (Pi) Model	$\vec{A} = \vec{D} = 1 + \vec{Z}_{TL} \vec{Y}_{TL}, \quad \vec{B} = \vec{Z}_{TL}, \quad \vec{C} = \vec{Y}_{TL} (2 + \vec{Z}_{TL} \vec{Y}_{TL})$	
Long Length Model	$\vec{A} = \vec{D} = \cosh(\vec{\gamma} \ell), \quad \vec{B} = \vec{Z}_c \sinh(\vec{\gamma} \ell), \quad \vec{C} = \vec{Z}_c^{-1} \sinh(\vec{\gamma} \ell)$ $\begin{cases} \vec{V}(x) = [\cosh(\vec{\gamma} x)] \vec{V}_R + [\vec{Z}_c \sinh(\vec{\gamma} x)] \vec{I}_R \\ \vec{I}(x) = [\vec{Z}_c^{-1} \sinh(\vec{\gamma} x)] \vec{V}_R + [\cosh(\vec{\gamma} x)] \vec{I}_R \end{cases}$ $x = 0$ on the sending end $x = \ell$ on the receiving end	